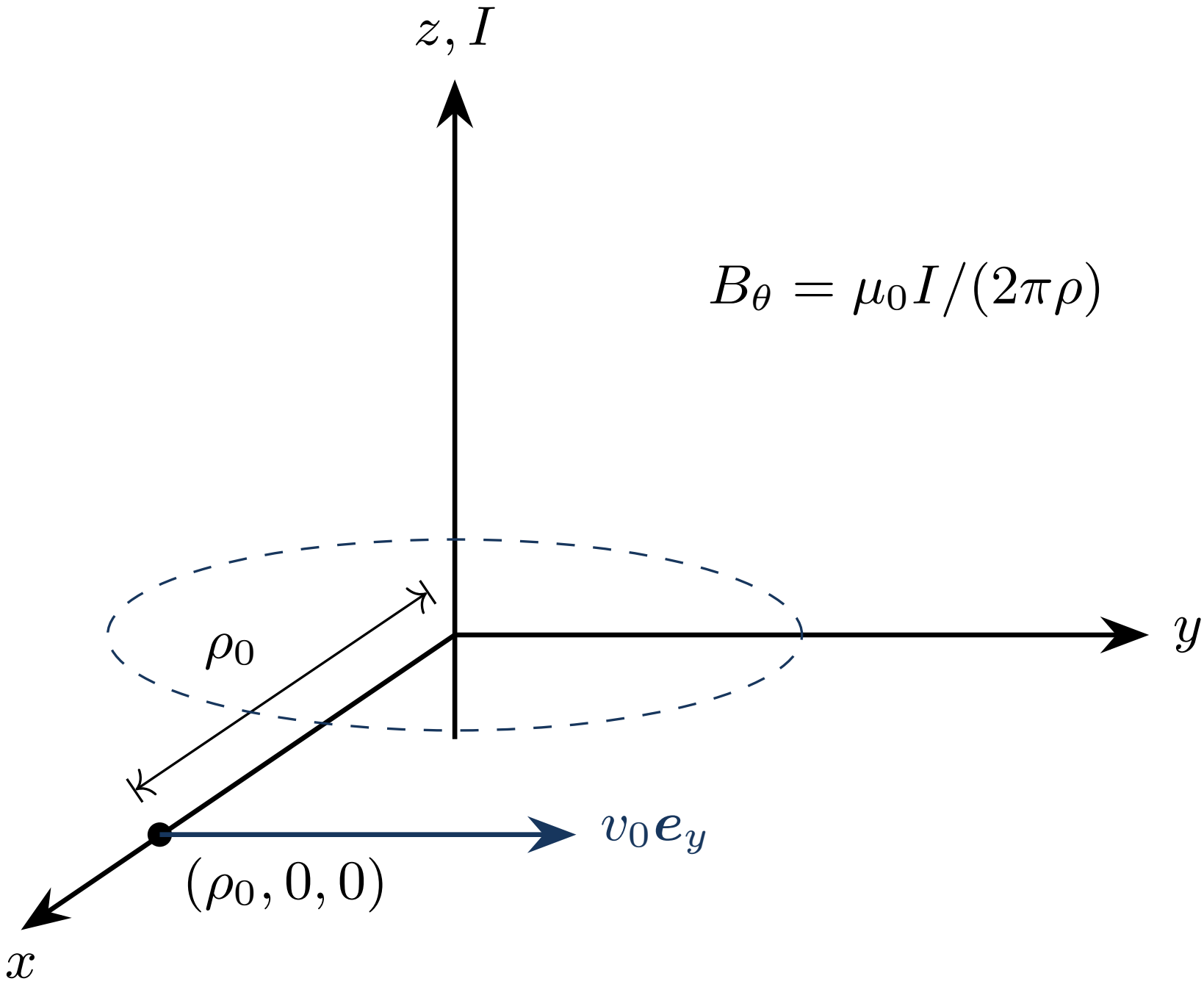
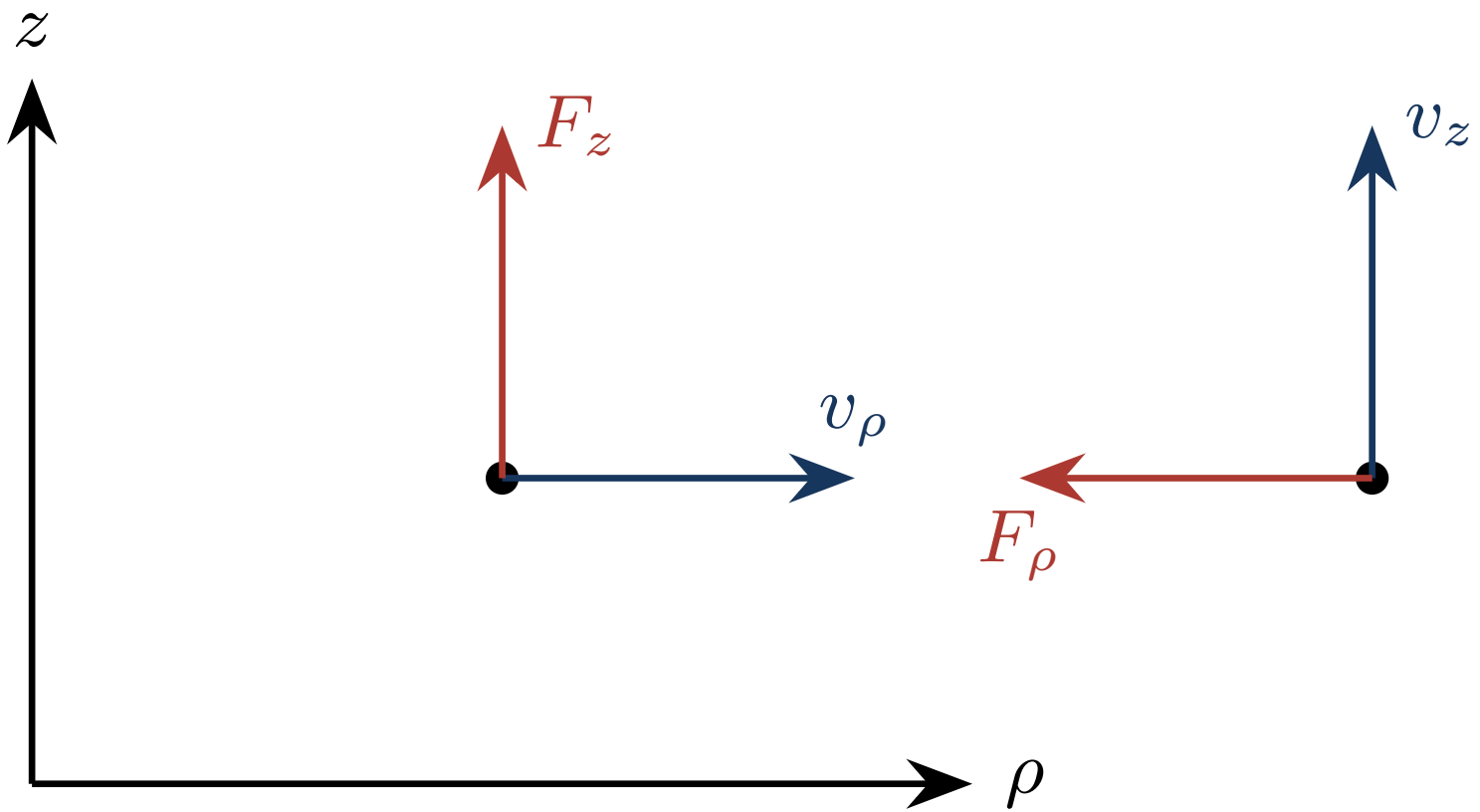
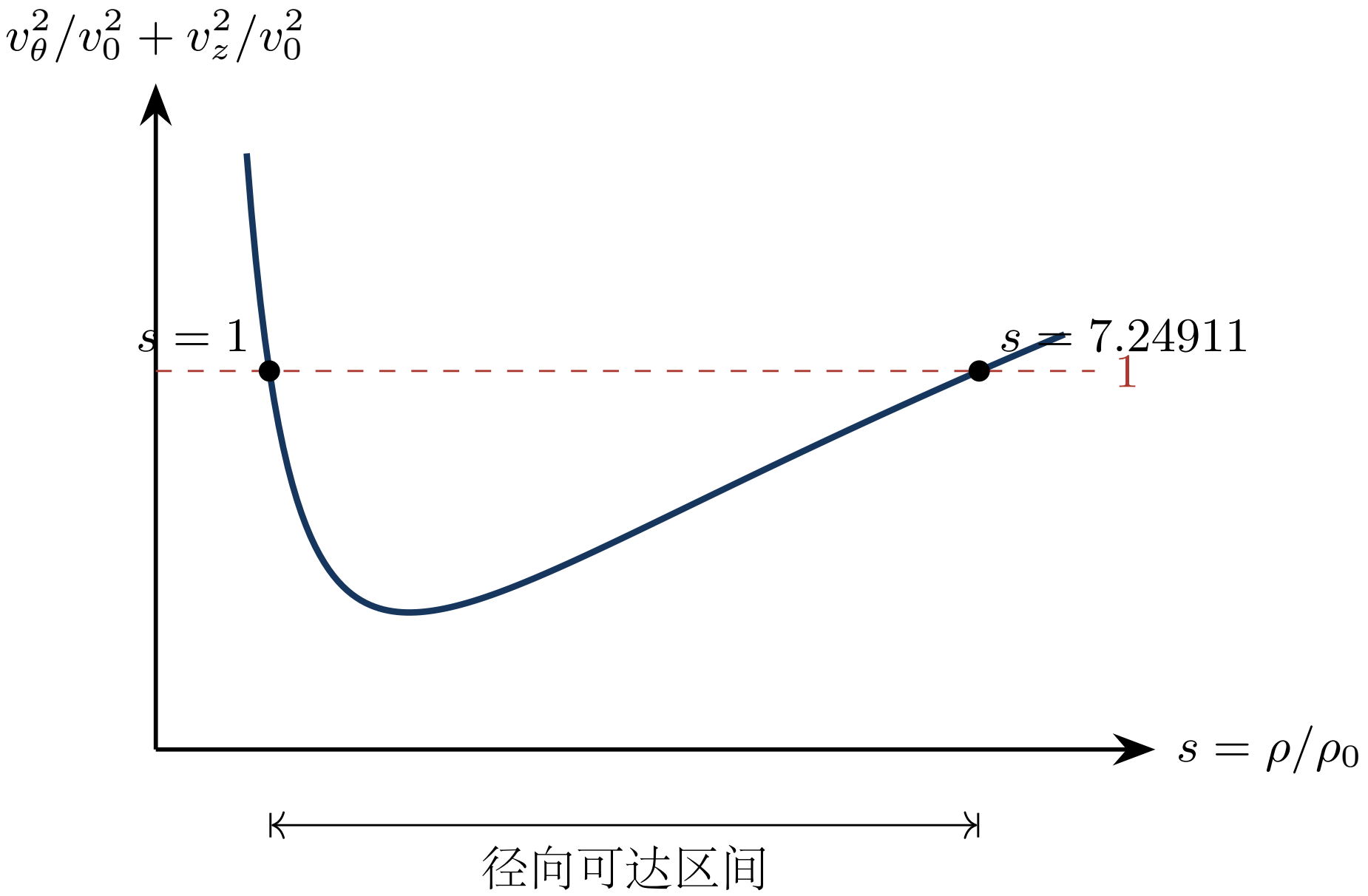

z, I
$B_\theta = \mu_0 I/(2\pi\rho)$
ρ_0
y
$v_0 \boldsymbol{e}_y$
$(\rho_0, 0, 0)$
x

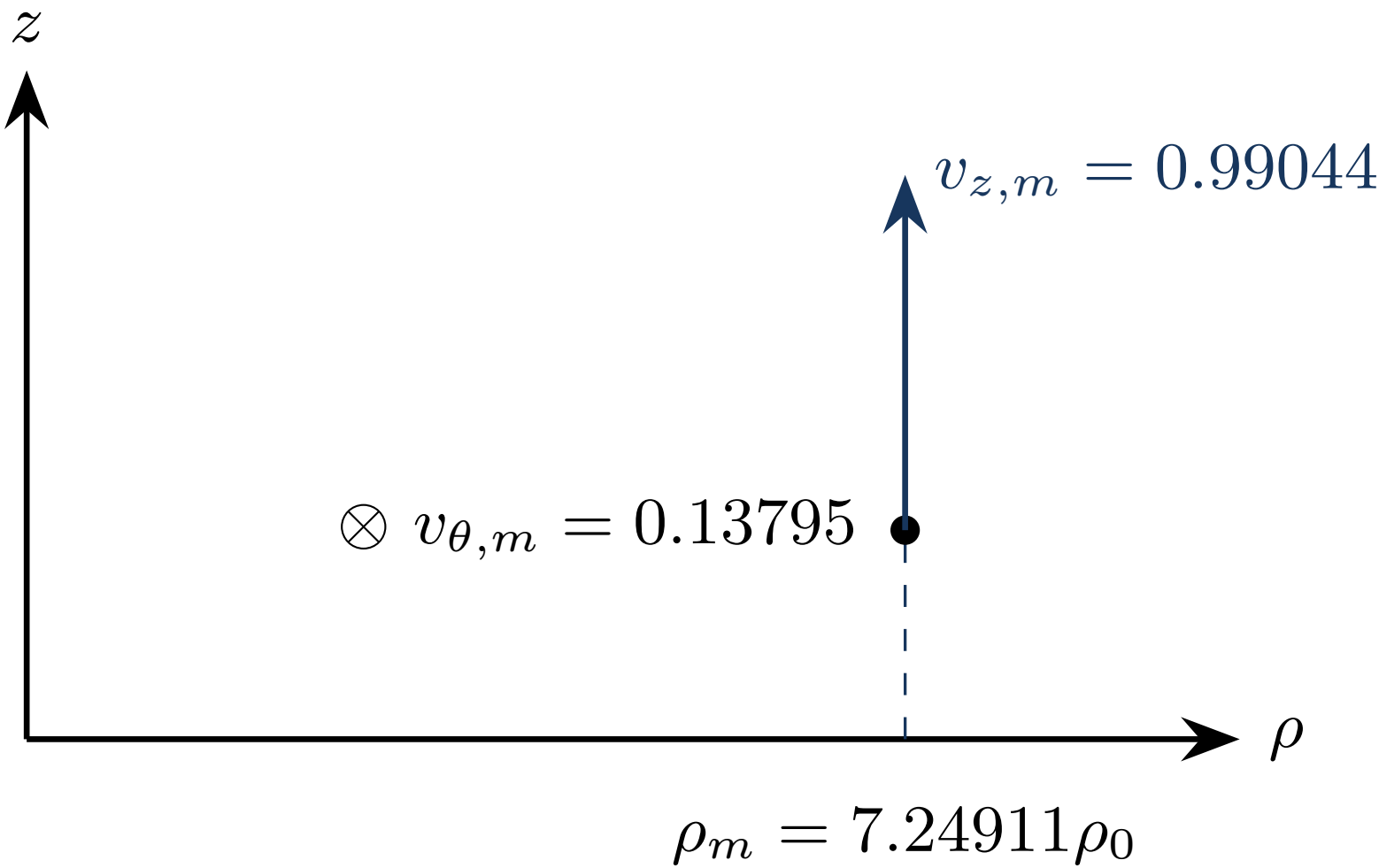
$$qI > 0, \;\; F_\theta = 0$$

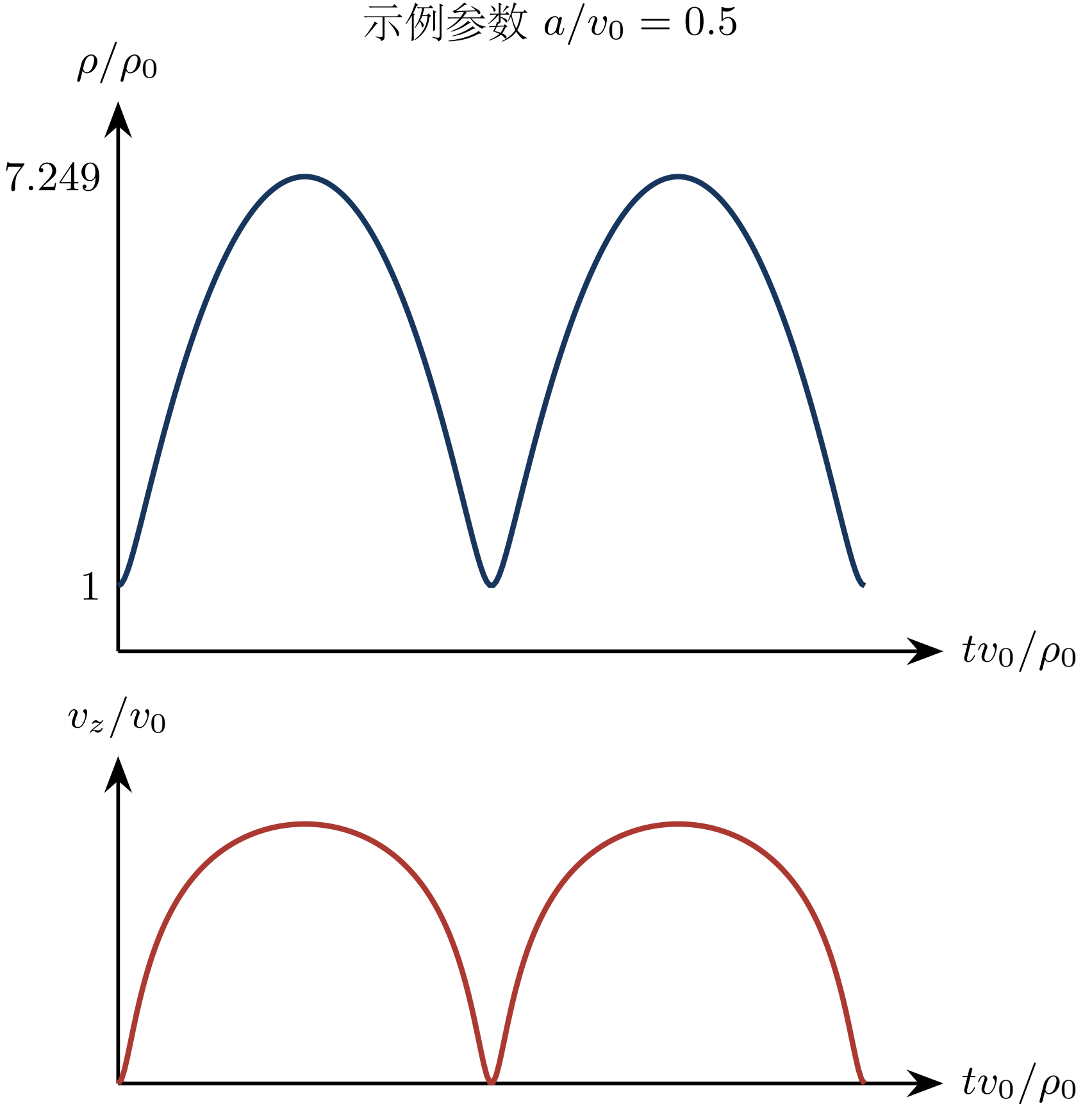


$$F_z = q\kappa v_\rho/\rho, \;\; F_\rho = -q\kappa v_z/\rho$$


$v_\theta^2/v_0^2 + v_z^2/v_0^2$
$s = 1$
$s = 7.24911$
1
$s = \rho/\rho_0$
径向可达区间

外转折处：$v_\rho = 0$，总速率仍为 1





两次径向周期